DO NOW — RECALL

# Processor and memory synoptic challenge

Name: _____ Class: _____ Date: _____

**1. What is an ADC used for?**

**2. Which register holds the current instruction?**

**3. What is the main limitation of two cores on wholly dependent steps?**

**4. A full adder receives 1,1 and carry-in 0. Give sum and carry-out and explain.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Converting an analogue sensor signal into a digital representation.

### 2. Model answer

CIR.

### 3. Model answer

The next step cannot start until the earlier result exists.

### 4. Model answer

$1+1+0=10_2$; sum bit 0, carry-out bit 1.

### Vocabulary — know these terms

PC — Program Counter; next instruction address.

MAR — Memory Address Register.

MDR — Memory Data Register.

cache — Fast nearby memory.

throughput — Work completed per unit time.

NEW LEARNING

# Read, annotate and explain

### Fetch the instruction

PC identifies address 72, which MAR sends to memory. MDR transfers its contents to CIR.

### Differentiate performance improvements

Cores can execute independent work; cache helps when needed data/instructions are present.

### Explain limits

Branching, shared memory bandwidth, serial dependencies and software constrain real speedup.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Starting PC=40, trace MAR, CIR and PC after a sequential fetch.

### Predict

Copy PC=40 into MAR to select the instruction location.

### Explain

Instruction contents travel via MDR to CIR, while sequential PC advances to 41.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

State what PC, MAR, MDR, CIR and ACC contain or do.

### Question 2

Explain how cache and extra cores could help video rendering, and why speedup may be less than expected.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Compare a multicore CPU and GPU for an interactive logic controller and a million independent pixels.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Doubling CPU cores always halves runtime.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.